



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS

9188/1

PAPER 1 Multiple Choice

NOVEMBER 2016 SESSION

1 hour

Additional materials:

Electronic calculator and/or Mathematical tables

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (Type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 14 printed pages and 2 blank pages.

Formulae

uniformly accelerated motion,	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on/by a gas,	$W = p\Delta V$
gravitational potential,	$\phi = -\frac{Gm}{r}$
refractive index,	$n = \frac{1}{\sin C}$
resistors in series,	$R = R_1 + R_2 + \dots$
resistors in parallel,	$1/R = 1/R_1 + 1/R_2 + \dots$
electric potential,	$V = \frac{Q}{4\pi\epsilon_0 r}$
capacitors in series,	$1/C = 1/C_1 + 1/C_2 + \dots$
capacitors in parallel,	$C = C_1 + C_2 + \dots$
energy of charged capacitor,	$W = \frac{1}{2}QV$
alternating current/voltage,	$x = x_0 \sin \omega t$
hydrostatic pressure,	$p = \rho gh$
pressure of an ideal gas,	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
radioactive decay,	$x = x_0 \exp(-\lambda t)$
decay constant,	$\lambda = \frac{0.693}{t_{1/2}}$
critical density of matter in the Universe,	$\rho_0 = \frac{3H_0^2}{8\pi G}$
equation of continuity,	$Av = \text{constant}$
Bernoulli equation (simplified),	$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$
Stokes' law,	$F = 6\pi\eta r v$
Reynolds' number,	$Re = \frac{\rho v r}{\eta}$
drag force in turbulent flow,	$F = Br^2 \rho v^2$

Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

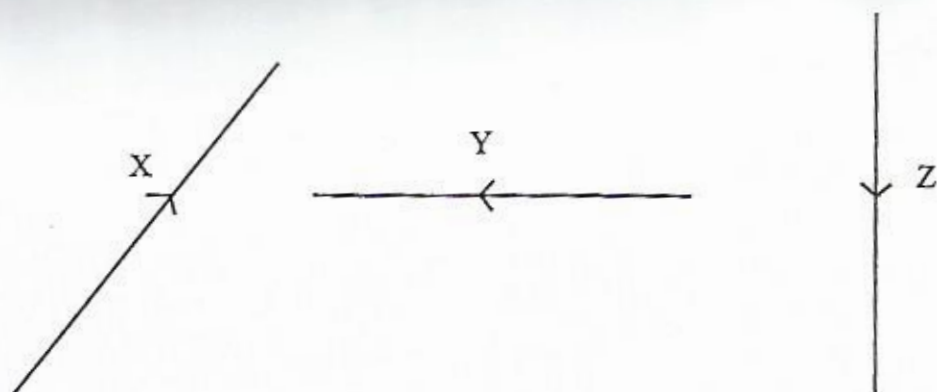
gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

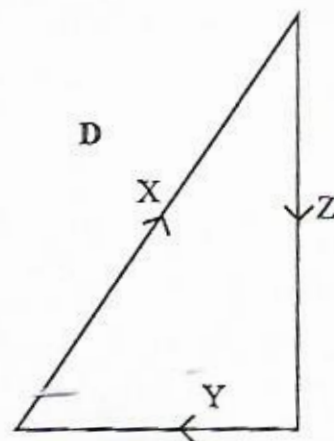
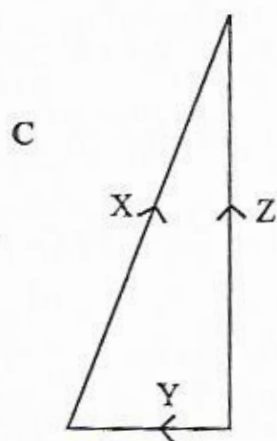
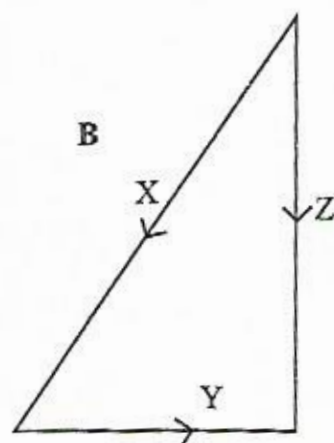
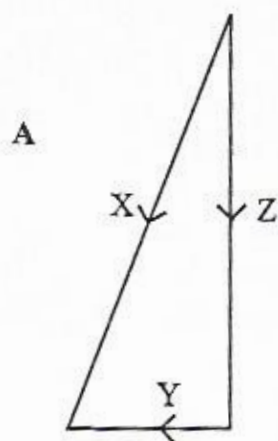
acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

1 The diagram shows three vectors X, Y and Z.



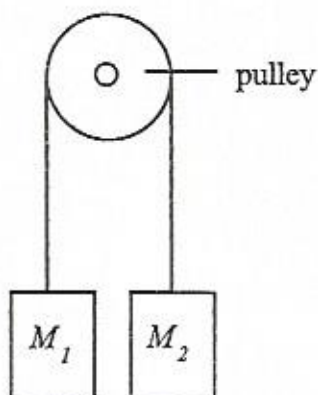
Which diagram is correct for the three vectors in equilibrium?



2 Which physical quantity is **not** basic?

- A mass
- B time
- C angle
- D amount of substance

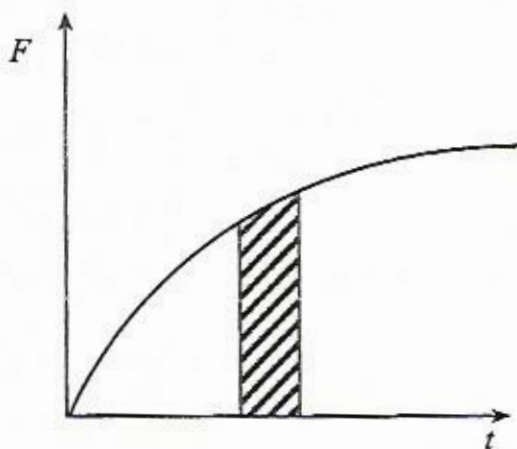
- 3 The diagram shows blocks of unequal masses, M_1 and M_2 , attached to the ends of a string passing over a frictionless pulley.



Which statement is true about the blocks after being released from rest?

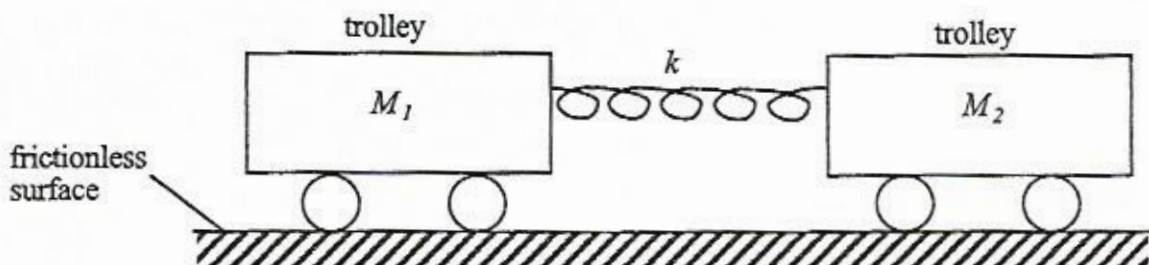
- A The total potential energy decreases while the total kinetic energy increases.
B The total potential energy increases while the total kinetic energy decreases.
C The total potential energy stays constant while total kinetic energy increases.
D The total potential energy increases while the total kinetic energy stays constant.
- 4 An object of mass, m , has momentum, p .
- The quantity $\frac{p^2}{m}$ has the same base units as
- A power.
B force.
C impulse.
D energy.
- 5 Which energy conversion, in a thermal power station, is **not** useful in the generation of electricity?
- A chemical to heat
B chemical to light
C heat to kinetic
D mechanical to electrical

- 6 The diagram shows the variation of force, F , t , acting on a car, with time, over a distance, s .



What quantity is represented by the shaded area?

- A gain in kinetic energy
 B loss in momentum
 C gain in momentum
 D loss in kinetic energy
- 7 The diagram shows two trolleys, of masses M_1 and M_2 on a frictionless surface, connected by a spring of spring constant, k . When compressed to a distance, x , and released, M_1 , has a velocity v_1 and mass M_2 has velocity v_2 .



Which row gives correct equations for the energy stored in the spring and the momentum of the trolleys on release?

	energy stored	momentum
A	$\frac{1}{2}kx^2$	$M_1v_1 + M_2v_2 = 0$
B	kx^2	$M_1v_1 = M_2v_2$
C	$\frac{1}{2}kx^2$	$M_1v_1 = M_2v_2$
D	kx^2	$M_1v_1 + M_2v_2 = 0$

- 8 The volume of a hot air balloon when fully inflated is $1.8 \times 10^3 \text{ m}^3$.

How much work is done in inflating the balloon?

[atmospheric pressure is 100 kPa].

- A 180 000 kJ
- B 180 kJ
- C 56 kJ
- D 56 J

- 9 Which wave quantity is **not** changed when a sound wave passes from air into water?

- A amplitude
- B frequency
- C velocity
- D wavelength

- 10 The minute hand of a wall clock is 14 cm long.

What is its average angular speed?

- A $2.4 \times 10^{-4} \text{ rads}^{-1}$
- B $1.7 \times 10^{-3} \text{ rads}^{-1}$
- C $1.0 \times 10^{-1} \text{ rads}^{-1}$
- D $6.3 \times 10^0 \text{ rads}^{-1}$

- 11 Which method is **not** used to remove accumulated dust on walls of the chimneys?

- A falling due to own weight
- B intermittent mechanical shaking
- C blowing of air through the chimney
- D switching off electric field between plates in the walls

- 12 The interference of monochromatic light produces dark and bright fringes on a screen. The dark fringes show absence of energy.

What happens to the energy for the dark fringes?

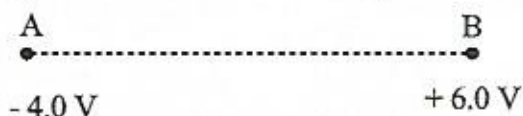
- A It is absorbed by the screen.
- B It is transmitted through the screen.
- C It is transferred to the bright fringes.
- D It is transferred to the surroundings.

- 13 A one atom thick film, was used in a diffraction experiment. Atoms in it were 10^{-10} m apart.

Which electromagnetic waves would cause the most significant diffraction pattern?

- A infrared
- B microwave
- C visible light
- D X-rays

- 14 The diagram shows two points A and B at different potentials.



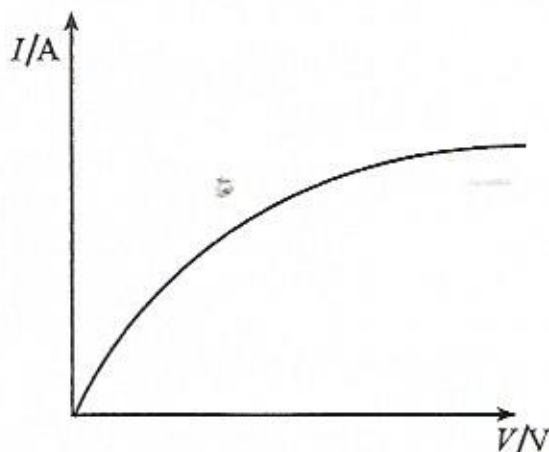
An electron passes point A with kinetic energy of 12.0 eV. When it reaches point B its kinetic energy will be

- A 2 eV.
 - B 6 eV.
 - C 18 eV.
 - D 22 eV.
- 15 In a double slit experiment, light of wavelength 600 nm gave a fringe separation of 0.4 mm on a suitably placed screen. Another monochromatic light source gave ten fringes across a distance of 3.3 mm on the same screen at the same position.

The wavelength of the second source was

- A 495 nm.
- B 660 nm.
- C 727 nm.
- D 7 300 nm.

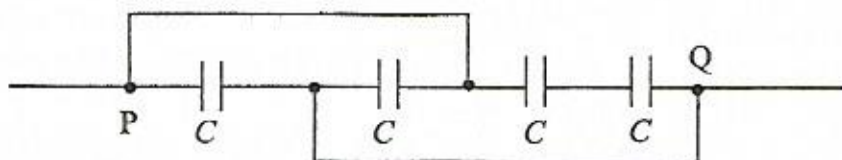
- 16 The diagram shows an I - V characteristic for an electrical component.



Which electrical component has this I - V characteristic?

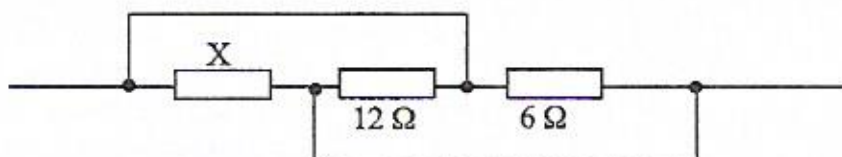
- A thermistor
- B filament lamp
- C diode
- D capacitor

- 17 The diagram shows identical capacitors, each of capacitance, C , connected together.



What is the effective capacitance between points P and Q?

- A $0.00\ C$
 B $0.25\ C$
 C $2.50\ C$
 D $4.00\ C$
- 18 The diagram shows a combination of resistors with a total resistance of $3\ \Omega$.

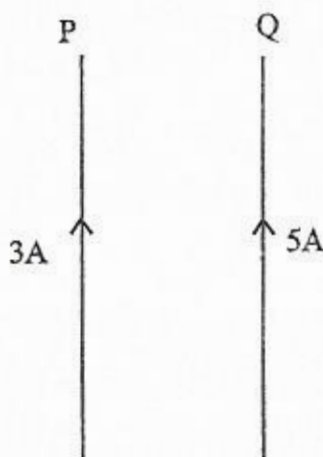


What is the resistance of X?

- A $2.0\ \Omega$
 B $3.6\ \Omega$
 C $6.0\ \Omega$
 D $12.0\ \Omega$
- 19 What is the order of magnitude of infra-red radiation wavelength?
- A $10^{-12}\ \text{m}$
 B $10^{-8}\ \text{m}$
 C $10^{-5}\ \text{m}$
 D $10^{-3}\ \text{m}$
- 20 A body experiences a force in a magnetic field when
- A charged and in motion.
 B uncharged and in motion.
 C charged and stationary.
 D uncharged and stationary.
- 21 Two identical point charges of charge $4.8 \times 10^{-19}\ \text{C}$ are placed $4.0\ \text{cm}$ apart in free space.
 What is the magnitude of the electric field strength at a point midway between the charges?

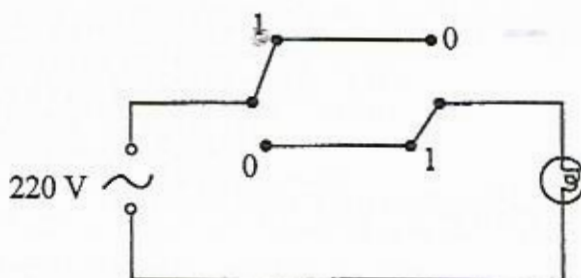
- A $0.0 \times 10^{-24}\ \text{N/C}$
 B $1.3 \times 10^{-24}\ \text{N/C}$
 C $1.1 \times 10^{-5}\ \text{N/C}$
 D $2.2 \times 10^{-5}\ \text{N/C}$

- 22 A charged capacitor stores energy because
- a power source is used to charge it.
 - the plates of a capacitor attract charge.
 - work is done to move charge from one plate to another.
 - there are positive and negative charges on each plate.
- 23 When the temperature of a conductor increases, its ability to conduct
- increases because the number of free electrons increases.
 - decreases because scattering of the free electrons by the metal ions increases.
 - increases because the mean time between the collisions of free electrons decreases.
 - decreases because the mean distance between the atoms increases.
- 24 The diagram shows two long parallel wires, P and Q. Each wire is carrying a current as shown.



Wire Q will experience a force directed

- away from P.
 - towards P.
 - out of the paper.
 - into the paper.
- 25 The diagram shows a stair lighting circuit.



Which single logic gate is demonstrated by the circuit?

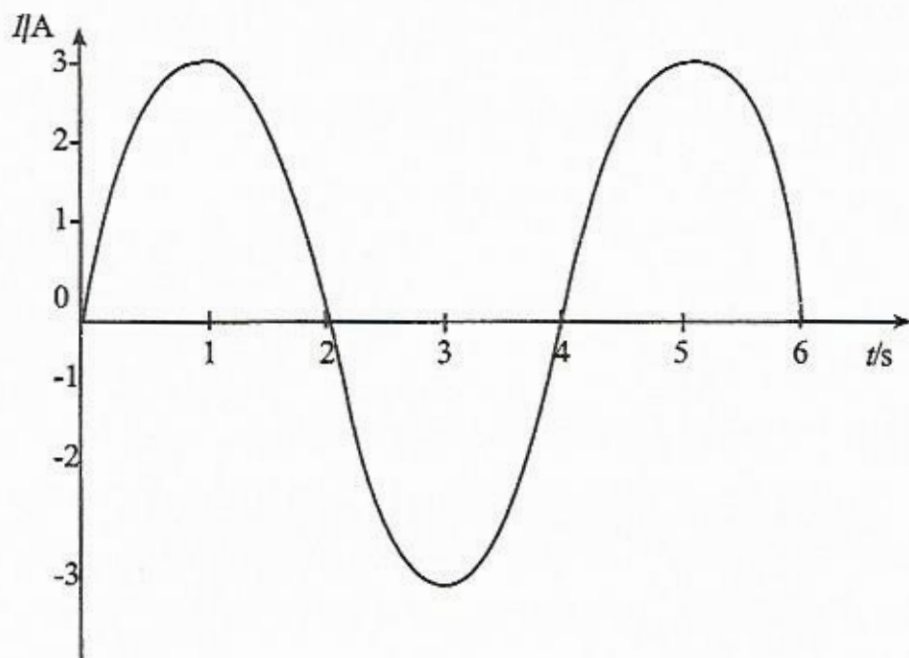
- AND
- EX-NOR
- EX-OR
- NOR

- 26 A constantan wire of length 3.0 m and a cross-sectional area of $2.95 \times 10^{-9} \text{ m}^2$ is used as a filament in a bulb connected to a power source of 220 V r.m.s. The peak current in the filament is 0.65 A.

What is the resistivity of the constantan wire?

- A $2.37 \times 10^{-7} \Omega/\text{m}$
- B $3.33 \times 10^{-7} \Omega/\text{m}$
- C $4.71 \times 10^{-7} \Omega/\text{m}$
- D $6.66 \times 10^{-7} \Omega/\text{m}$

- 27 The graph shows alternating current varying with time.



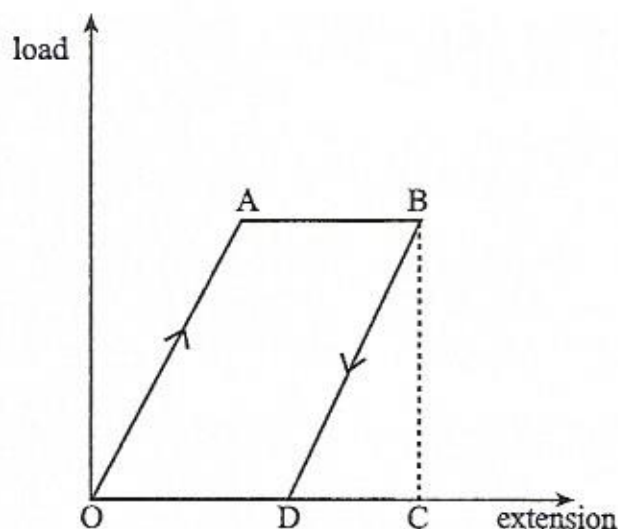
What is the r.m.s value?

- A 1.5 A
- B 2.1 A
- C 3.0 A
- D 4.5 A

- 28 Which list consists of output transducers only?

- A microphone, loudspeaker and headphone
- B keyboard, microphone and monitor
- C transistor, light emitting diode and light dependent resistor
- D loudspeaker, cathode ray oscilloscope and headphone

- 29 The diagram shows a load-extension graph for a stretched wire.



What is represented by areas OABCO and OABDO?

	area OABCO	area OABDO
A	lost energy	total strain energy
B	recovered energy	lost energy
C	recovered energy	total strain energy
D	total strain energy	lost energy

- 30 What is the effect of negative feedback on the gain and bandwidth of an operational amplifier?

	effect on gain	effect on bandwidth
A	reduced	increased
B	reduced	reduced
C	increased	reduced
D	increased	increased

- 31 Two molecules of a gas have speeds 2 kms^{-1} and 8 kms^{-1} .

What is the root-mean-square speed of the molecules?

- A $\sqrt{5} \text{ kms}^{-1}$
 B $\sqrt{10} \text{ kms}^{-1}$
 C $\sqrt{34} \text{ kms}^{-1}$
 D $\sqrt{68} \text{ kms}^{-1}$

- 32 Which is **not** an assumption for the kinetic theory of gases?

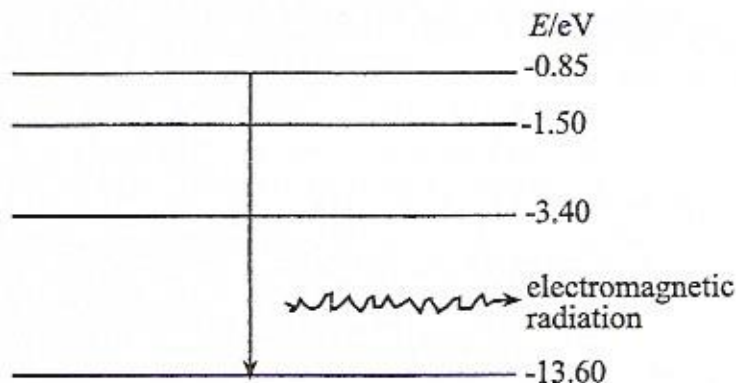
- A The attraction between molecules is negligible.
 B The volume of a molecule is negligible compared to the volume occupied by the gas.
 C The time of collision is negligible compared to the time between collisions.
 D The molecules are perfectly elastic spheres and have random motion.

- 33 If a substance is heated, the
- A internal energy of the substance rises.
 - B internal energy of the substance drops.
 - C total kinetic energy of its particles increases.
 - D total kinetic energy of its particles decreases.
- 34 An electron volt is a unit used to measure
- A electric field.
 - B potential difference.
 - C force.
 - D energy.
- 35 The properties of photoelectricity include:
1. instantaneous emission of photoelectrons
 2. increase in photocurrent when the intensity of the incident wave is increased
 3. the variation of the kinetic energy from zero to a maximum

Which of the three properties are evidence for the particulate nature of electromagnetic waves?

- A 1 and 2
 - B 1 and 3
 - C 2 and 3
 - D 1, 2 and 3
- 36 The energy of a photon entering a glass block from a vacuum
- A stays the same because its speed remains constant.
 - B stays the same because its frequency remains constant.
 - C increases because the wavelength increases.
 - D decreases because the wavelength decreases.

- 37 Electromagnetic radiation is emitted from an excited atom as shown.



In which region of the electromagnetic spectrum does the radiation belong?

- A visible region
 B ultra-violet region
 C gamma region
 D X-ray region
- 38 A certain radioactive isotope has a decay constant of $1.5 \times 10^{-3} \text{ s}^{-1}$. In a sample, it is discovered that only 30% of the undecayed isotope remains.
- What time has passed since the initial measurement?
- A 191 s
 B 227 s
 C 445 s
 D 766 s
- 39 What is observed in the α -particle scattering experiment?
- A A few α -particles were deflected through angles greater than a right angle.
 B All α -particles were deflected through angles greater than a right angle.
 C Most α -particles were deflected through angles greater than a right angle.
 D No α -particles were deflected through an angle greater than a right angle.
- 40 The half-life of a certain radioactive element is such that $\frac{5}{8}$ of a given quantity decays in 20 days.

How long does it take for the quantity to decay to $\frac{1}{64}$ of the original quantity?

- A 0.5 days
 B 85 days
 C 176 days
 D 800 days

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS**9188/2**

PAPER 2 Theory

NOVEMBER 2016 SESSION**1 hour 15 minutes**

Candidates answer on the question paper.

Additional materials:

Electronic calculator and/or Mathematical tables

TIME 1 hour 15 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

For numerical answers, **all** working should be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	

This question paper consists of 12 printed pages.

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Turn over

Answer *all* questions.

- 1 (a) Distinguish between a *vector quantity* and a *scalar quantity*.

[1]

- (b) Fig. 1.1 shows a system of forces acting at a point.

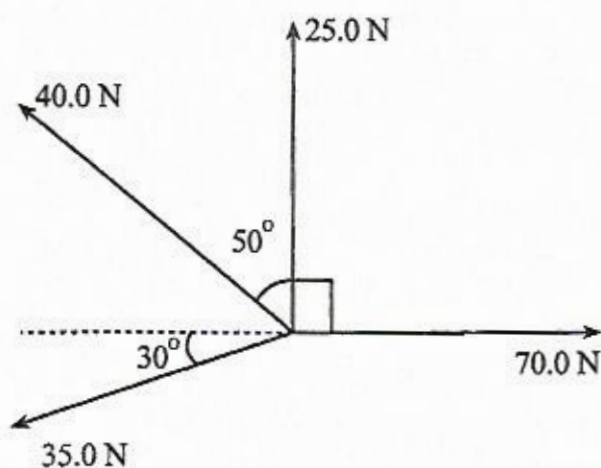


Fig. 1.1

Calculate the magnitude of the resultant force.

magnitude of resultant force = _____ [3]

- 2 (a) State the principle of conservation of momentum.

[2]

- (b) Fig. 2.1 shows a snooker ball of mass 0.40 kg travelling at 2.0 m/s colliding head on with another identical ball travelling at 1.0 m/s . After the collision, the first snooker ball bounces back at 0.50 m/s .

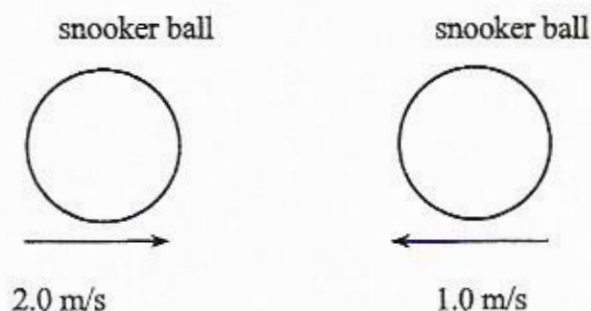


Fig. 2.1

- (i) Show that the second ball bounces back as well.
- (ii) Determine whether the collision is elastic or inelastic.

[5]

- (c) Explain why a child jumping off a wall should bend knees on landing.

[2]

- 3 (i) State the **two** conditions that must be satisfied for a satellite to be in a geostationary orbit.

1.

2.

- (ii) The mass and the radius of the earth are 6.02×10^{24} kg and 6.4×10^6 m respectively.

Calculate the height above the earth's surface of the satellite in (i).

height of satellite =

[5]

- 4 (a) Define *electric field strength*.

[1]

- (b) An oil drop of weight of 3.04×10^{-14} N is held stationary between metal plates 4.0 mm apart when the potential difference applied is 380 V.

- (i) Calculate the

1. electric field strength, E , between the plates,

$$E = \underline{\hspace{2cm}}$$

2. magnitude of the charge on the oil drop.

$$\text{charge} = \underline{\hspace{2cm}}$$

- (ii) Hence show that the charge is quantized.

[6]

5 (a) Explain the terms with reference to an op-amp:

(i) *infinite slew-rate*

(ii) *infinite bandwidth*

(iii) *infinite input impedance*

(iv) *zero output impedance*

[4]

(b) Fig. 5.1 shows an inverting amplifier.

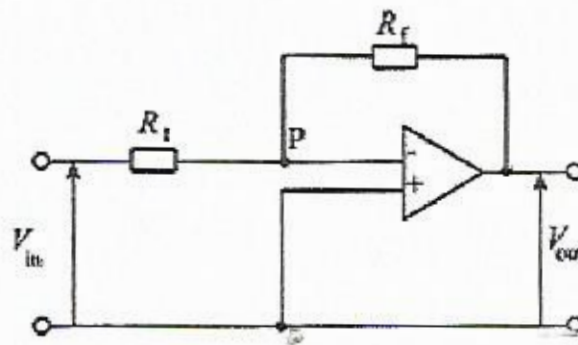


Fig. 5.1

Derive an expression for the gain, A , of the amplifier.

[2]

6 (a) Define

(i) *yield point*,

(ii) *ultimate tensile strength*.

[2]

(b) In an experiment to measure the Young Modulus of a wire, the wire is long and thin.

Explain why the wire should be

(i) *long*, _____

(ii) *thin*, _____

[2]

(c) When a load of 40.0 N is applied to a wire of cross-sectional area $1.50 \times 10^{-7} \text{ m}^2$ and length 2.00 m, it causes an extension of 1.00 mm.

Determine the

(i) stress in the wire,

(ii) energy stored in the wire.

[3]

- 7 (a) (i) Define *thermal equilibrium*.

- (ii) Explain the importance of thermal equilibrium.

[3]

- (b) A resistance thermometer has resistances of $37.3 \, \Omega$ and $48.1 \, \Omega$ at ice and steam points respectively.

Calculate the temperature corresponding to $18.2 \, \Omega$.

temperature = _____ [2]

- 8 (a) State the ideal gas equation.

[1]

- (b) The density of nitrogen at standard temperature and pressure (s.t.p) is $1.25 \, \text{kgm}^{-3}$. The s.t.p values are $1.01 \times 10^5 \, \text{Pa}$ and $273 \, \text{K}$.

Calculate the r.m.s speed, $c_{r.m.s}$, of nitrogen molecules at $298 \, \text{K}$.

$c_{r.m.s} =$ _____ [3]

- (c) Explain how an increase in internal energy of an ideal gas is related to a rise in temperature of the ideal gas.

[2]

- 9 (a) Given the photoelectric equation:

$$hf = \phi + E_k$$

State the physical quantity represented by

- (i) hf , _____
- (ii) ϕ , _____
- (iii) E_k , _____

[3]

- (b) A metal surface of threshold wavelength 550 nm is irradiated with blue light of wavelength 450 nm.

Calculate the

- (i) maximum kinetic energy of the emitted photoelectrons,
- (ii) reverse voltage required to stop the emitted photoelectrons from reaching the anode.

[4]

- (c) State, with a reason, the effect on the quantity calculated in b(ii) of doubling the

(i) intensity of incident light, _____

(ii) wavelength. _____

[4]



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS
PAPER 3 Theory

9188/3

NOVEMBER 2016 SESSION

50 minutes

Additional materials:

Answer paper

Electronic Calculator and / or Mathematical tables

Ruler (mm)

TIME 50 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **three** questions.

Question 1 is compulsory.

Answer any other **two** from the remaining questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

All working for numerical answers must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question. You are reminded of the need for good English and clear presentation in your answers.

Candidates are advised to spend 25 minutes on **question 1**.

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Answer **question 1** and **any other 2** from the remaining questions.

- 1 (a) **Fig. 1.1** shows a vernier callipers before and after making measurements of the internal diameter of a cylindrical tube. The tube has a length of 25.0 cm as measured by a ruler with an accuracy of 1 mm.



Fig. 1.1

- (i) Write the readings shown in **Fig. 1.1**.
- (ii) Calculate, with its uncertainty, the internal
 1. diameter of the tube,
 2. volume of the tube.

[6]

- (b) (i) Distinguish between *precision* and *accuracy*.
- (ii) Two sets of readings, **A** and **B**, were obtained by two students in experiments to determine the acceleration due to gravity.

[All readings are in ms^{-2} .]

Set A: 7.88; 7.87; 7.86

Set B: 8.75; 10.23; 10.45

State whether or **not** the sets of values are precise and accurate.

[5]

- (c) (i) State *the principle of superposition*.
- (ii) Fig. 1.2 shows a pair of loudspeakers 1.5 m apart and 7 m away from the line AYZ . The loudspeakers are emitting sound of frequency 1 320 Hz, and speed of 330 m/s. A listener moving from A to Z hears a minimum sound at A and Z, and a maximum sound at Y.

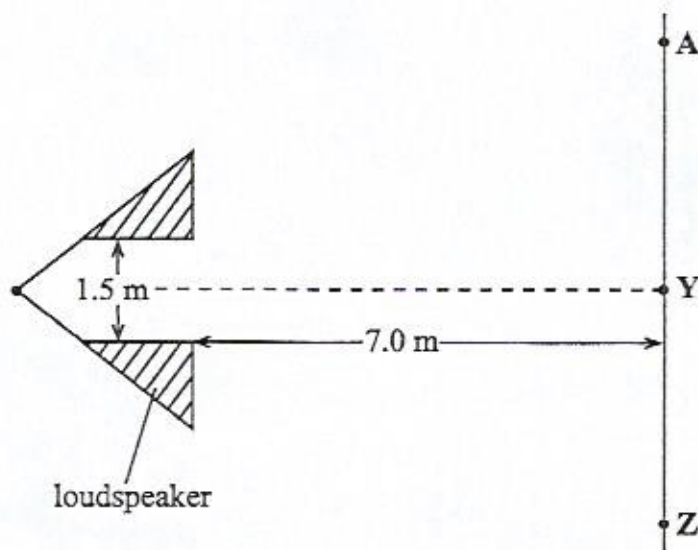


Fig. 1.2

Calculate the distance between A and Z.

[5]

- (d) Fig. 1.3 shows a model of a front-end loader with a square base of side 1.2 m used to demonstrate the moment of a force. A uniform lever, with first and second arms each of length 1.5 m and weight 20 N, is pinned at the mid-point of the base.

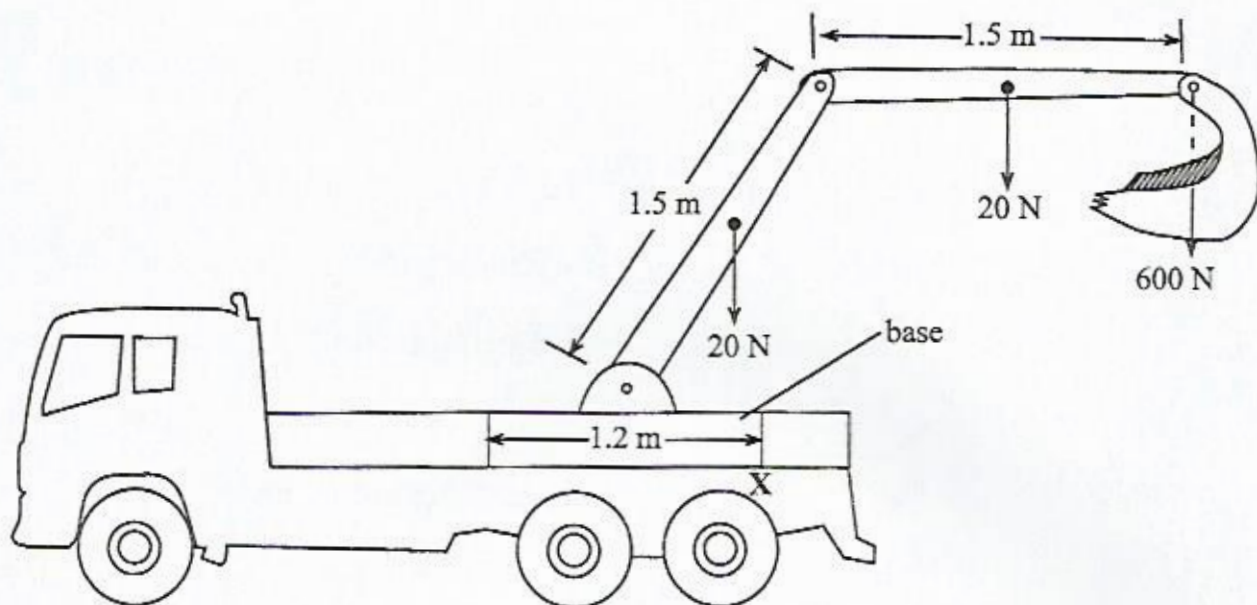


Fig. 1.3

The base is designed such that there is no toppling over at X when the arm is fully extended.

- (i) Explain the term *moment of a force*.
- (ii) Find the minimum weight of the base that just prevents toppling over at point X when the arm is fully extended.

[4]

- 2 (a) An equation for a progressive wave in a string is given by the expression:

$$Y = 2 \times 10^{-3} \sin[600t - 20X]$$

A series of such waves are transmitted along a stretched string fixed at one end. The waves reflected at the fixed end without loss of energy and superposition occurs in the string.

- (i) Write an equation for the reflected wave.
- (ii) Describe how these waves satisfy conditions for the production of a stationary wave.
- (iii) Derive an equation for the resultant wave using the principle of superposition.

[5]

- (b) A damping material is placed at the fixed end of the string in 2(a) so that the incident progressive waves are absorbed.

Find the

- (i) wavelength of the progressive wave in a(i),
 (ii) maximum speed of a particle in the string.

[4]

- (c) Explain why an external agent is always required to maintain a stationary wave in a stretched string.

[1]

- 3 (a) (i) Define *impulse*.

- (ii) State the SI unit of impulse.

[2]

- (b) A 7 500 kg lorry travelling at 5 m/s due east collides with a 1 500 kg car moving at 20 m/s in the direction South 60° West. The two vehicles coalesce.

Determine the

- (i) direction and

- (ii) speed

of the wreckage.

[6]

- (c) Fig. 3.1 shows some of the safety features in a car which include safety belt and head restraint. Explain how the head restraint prevents the driver from injury if the car is hit from behind.

[2]

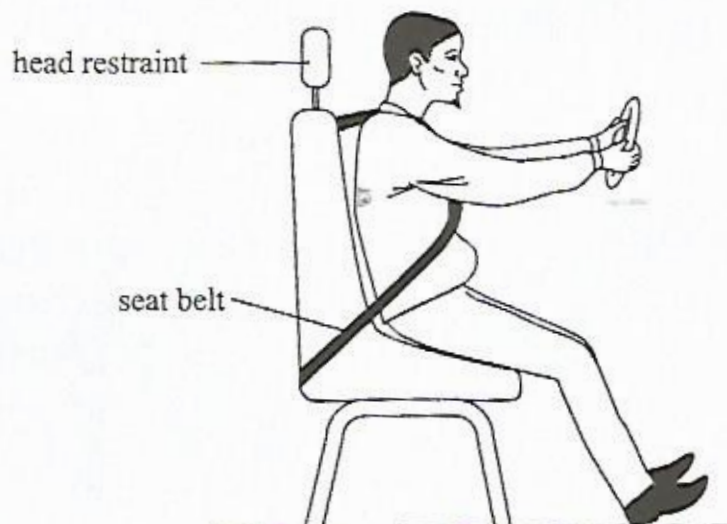


Fig. 3.1

4 (a) Define the

(i) *gravitational potential* and

(ii) *gravitational field strength*.

at a point in a gravitational field.

[2]

(b) (i) State the significance of

1. the negative sign in the equation for gravitational potential,

2. a test mass in the study of the gravitational field.

(ii) Show that the kinetic energy, E_k , of a satellite of mass, M , orbiting the Earth of mass, M_E , at a radius R , is given by $E_k = \frac{GM_E M}{2R}$.

[6]

(c) The satellite in (b)(ii) speeds up when it falls to a lower orbit, hence its kinetic energy increases.

Explain why the satellite loses energy.

[2]



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

9188/4

PAPER 4 Practical Test

NOVEMBER 2016 SESSION

2 hours 30 minutes

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Electronic calculator

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page and on any separate answer paper used.

Answer all questions.

Write your answers in the spaces provided on the question paper.

In Questions 1 and 2, you are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them. The working of the answers is to be handed in. Marks are mainly given for a clear record of the observations actually made, for their suitability and accuracy, and for the use made of them. **Routine** precautions and theory are **not** wanted in Questions 1 and 2. You should, however, record any **special** precautions you have taken so as to aid accuracy. At the end of the examination, fasten any separate answer paper used securely to the question paper.

INFORMATION FOR CANDIDATES

Questions 1 and 2 carry 18 marks each and question 3 carries 14 marks.

Squared paper and Mathematical tables are available.

Additional paper and graphs should be submitted **only** if it becomes **necessary** to do so.

You are advised to spend approximately one hour on each of Questions 1 and 2 and 30 minutes on Question 3

You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE

1	
2	
3	
TOTAL	

This question paper consists of 8 printed pages, 2 lined pages and 2 blank pages.

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It is recommended that you spend about 60 minutes on this question.

- 1 In this experiment you will investigate the variation of the extension, x , of springs connected in series with the original length, L_o , of the springs.

- (a) (i) Measure and record the original length, L_o , of one spring.
 (ii) Set up the apparatus as shown in Fig. 1.1.

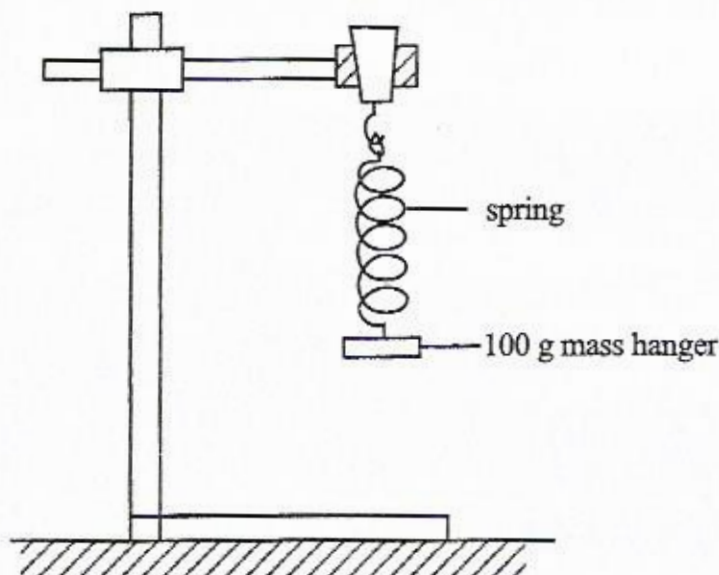


Fig. 1.1

- (iii) Measure and record the extension, x , of the spring when it is loaded with a 100 g mass hanger.
- (b) (i) Change L_o by connecting another spring in series to the one in a(i). Repeat a(iii).
 (ii) Repeat b(i) to obtain **four** further measurements by connecting more springs in series.

In your table of results include values of $\log L_o$ and $\log x$.

- (iii) Justify the number of significant figures you have given for $\log L_o$.
- (c) Theory suggests that L_o and x are related by the equation $L_o = qx^n$, where q and n are constants.
 (i) Plot a graph of $\log L_o$ (y-axis) against $\log x$ (x-axis).
 (ii) Use the graph to determine the values of q and n .
- (d) Suggest a possible way of improving the accuracy of the results.

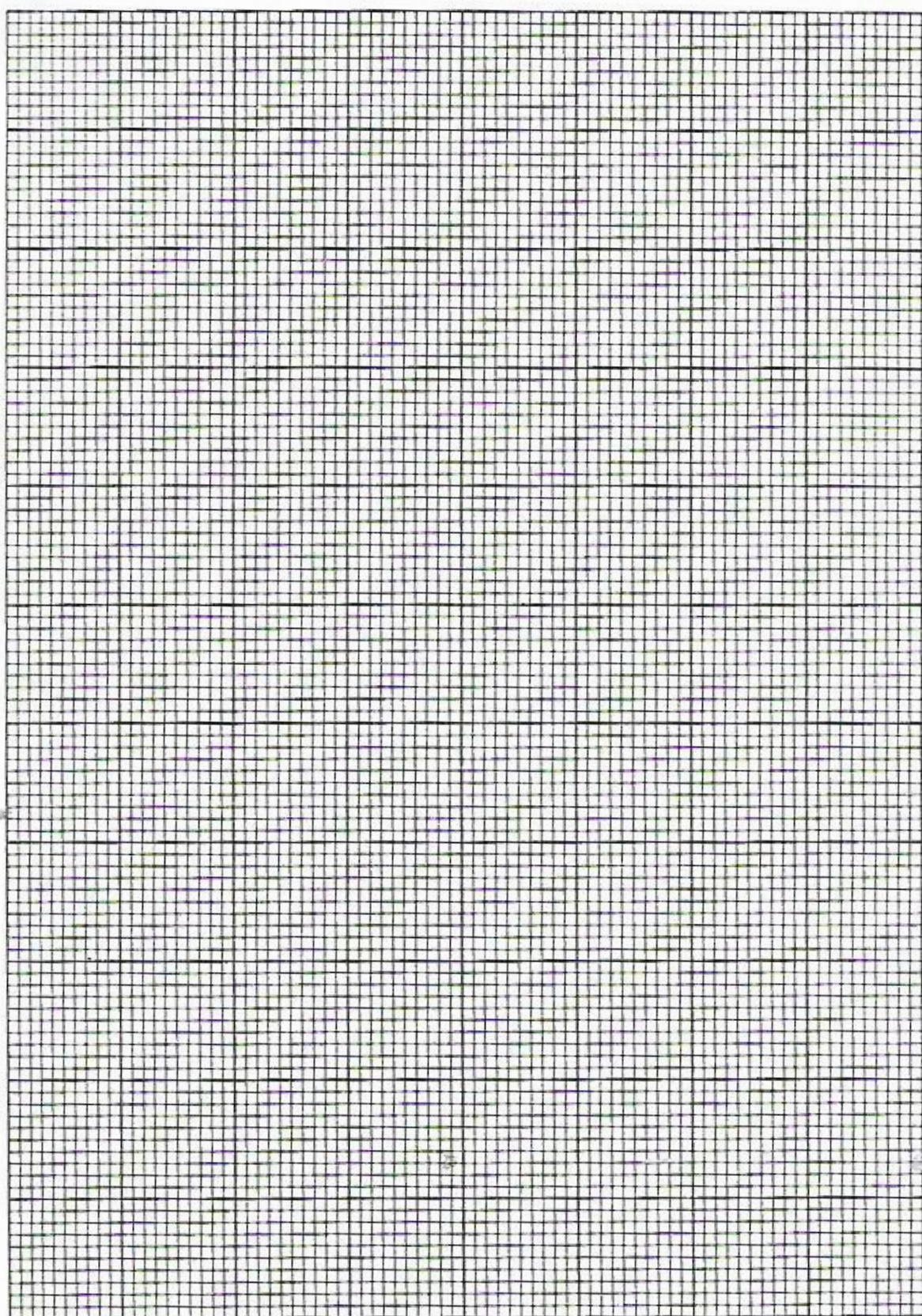
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Measurements and calculations

M

R

A



G

It is recommended that you spend about 60 minutes on this question.

- 2 In this experiment you will investigate the variation of the output voltage, V , from an operational amplifier with the feedback resistance, R_f .

(a) The circuit in Fig. 2.1 has been set up for you with $R_f = 50\ \Omega$.

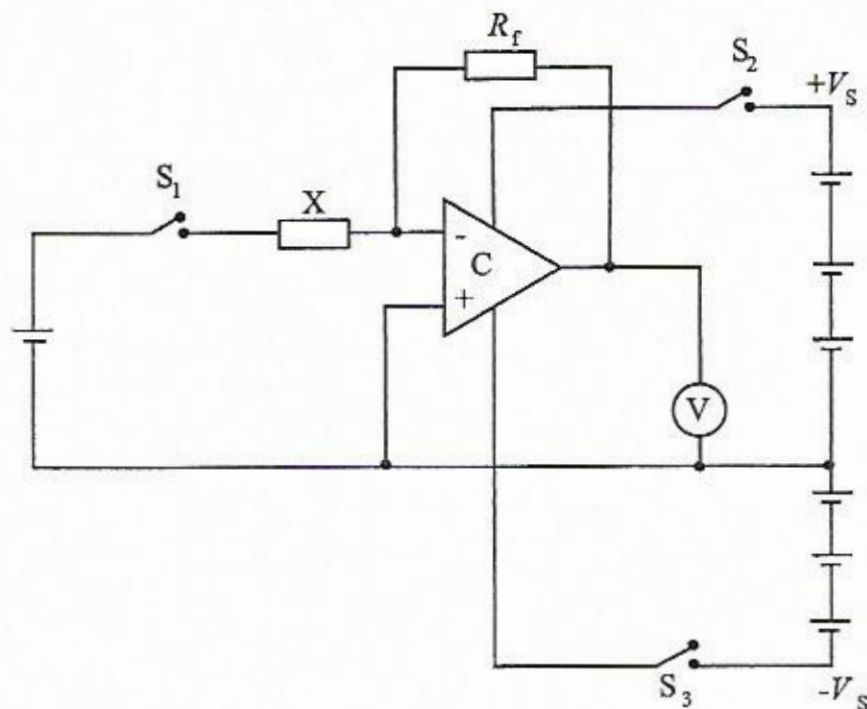


Fig. 2.1

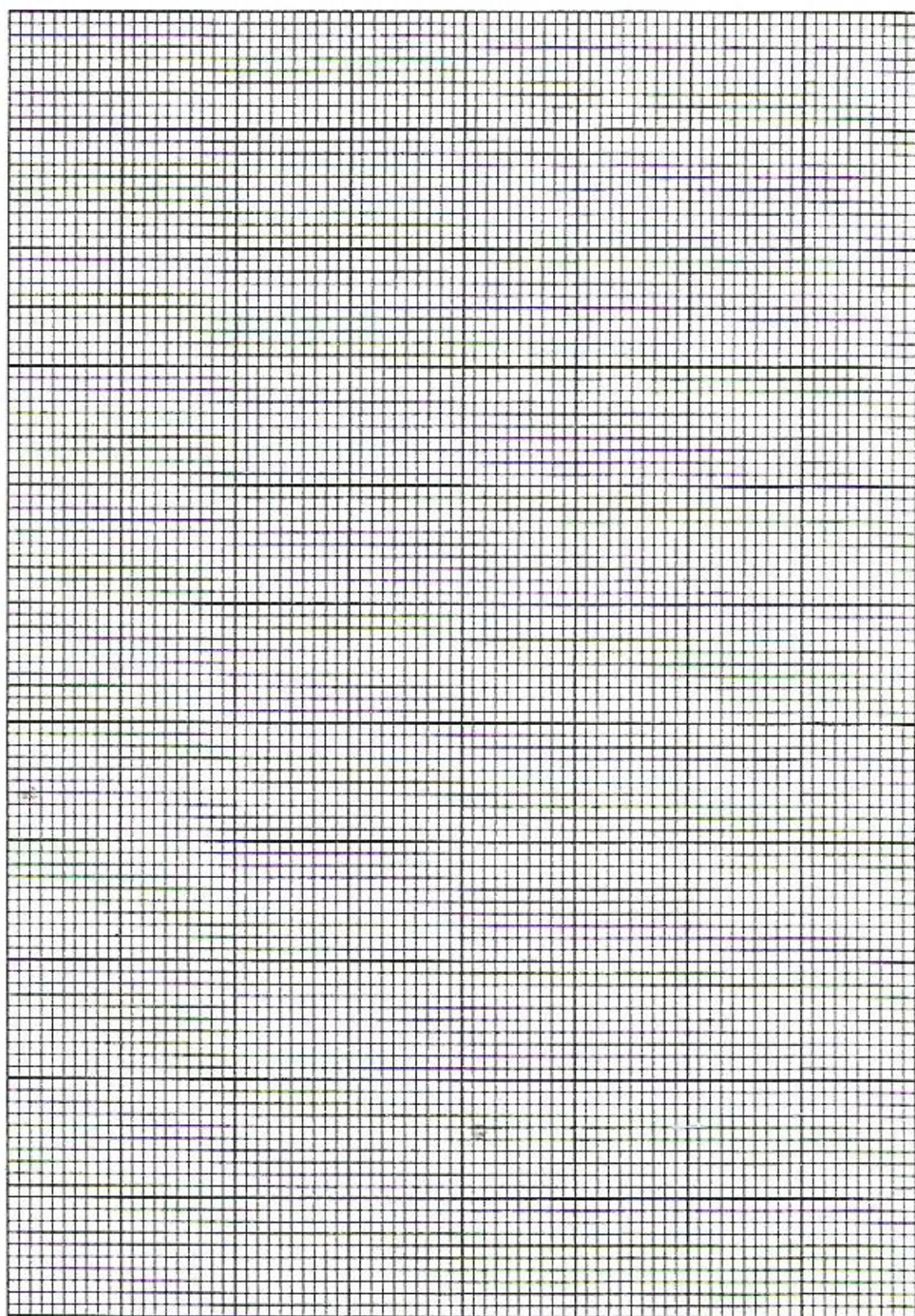
- (b) (i) Close the three switches.
Record the voltmeter reading, V and the value of R_f in your table of results.
Open the switches.
- (ii) Change R_f by connecting another resistor in series and repeat b(i).
- (iii) Repeat b(ii) for R_f in the range $50\ \Omega \leq R_f \leq 1200\ \Omega$ for at least 13 sets of values of R_f and V .
- (c) (i) Plot a graph of V (y-axis) against R_f (x-axis).
(ii) Describe the trend of the graph.
- (d) The feedback current is given by the gradient of the graph in c(i).
- (i) Determine the feedback current when the resistance R_f is $450\ \Omega$.
- (ii) Justify the number of significant figures used for the value in d(i).

Measurements and calculations

M

R

A



It is recommended that you spend about 30 minutes on this question.

- 3 When excessive current flows through a fuse the fuse wire becomes hot and it melts.

A student suggests that the temperature of a wire carrying a current depends on the

- (i) current value,
- (ii) the diameter and
- (iii) length of the wire.

Design a laboratory experiment to investigate the student's suggestion.

In your account you should pay particular attention to:

the procedure
control of variables
precautions

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ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS
PAPER 5 Theory

9188/5

NOVEMBER 2016 SESSION

1 hour 15 minutes

Additional materials:

Answer paper

Electronic Calculator and / or Mathematical tables

Ruler (mm)

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **four** questions.

Question 1 is compulsory.

Answer any other **three** from the remaining questions.

Write your answers on the separate answer paper provided. If you use more than one sheet of paper, fasten the sheets together. All working for numerical answers must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question. You are reminded of the need for good English and clear presentation in your answers.

Candidates are advised to spend 30 minutes on **question 1**.

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Answer question 1 and any other 3 from the remaining questions.

- 1 (a) (i) State **one** practical application of the electrostatic phenomena.
- (ii) Describe the construction of a simple lightning conductor.
- (iii) Suggest a reason why a human hand can be used to discharge a charged metal rod.

[4]

- (b) (i) Define *magnetic flux density*.
- (ii) Fig. 1.1 shows a straight wire XY, 10 cm long, perpendicular to a magnetic field between two permanent magnets.

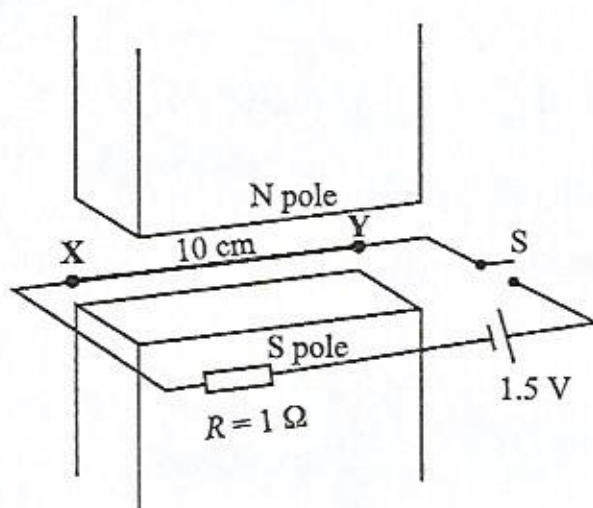


Fig. 1.1

State and explain the observations expected when switch S is closed.

- (iii) Calculate the force experienced by the wire, XY, if the magnetic flux density is $2 \times 10^{-3} \text{ T}$.

[6]

- (c) (i) Explain the term
1. *creep*,
 2. *fatigue*.
- (ii) Describe **one** practical situation that may lead to fatigue failure.
- (iii) Suggest, with a reason, **one** property of a material suitable to make a tractor trailer.

[5]

- (d) The work function, ϕ , of a metal is given by the equation

$$\phi = hf - \frac{1}{2}mv_{\max}^2.$$

- (i) State the significance of $v_{\max}$.
- (ii) A metal surface having a work function of 3.03 eV is illuminated with radiation of wavelength 350 nm.

Calculate the maximum kinetic energy of the emitted photoelectrons.

- (iii) Explain why some electrons do not attain the kinetic energy in (ii).

[5]

- (e) (i) Define the *half-life* of a radioactive substance.

- (ii) The half life of copper-64 is 13.0 hours.

Find the percentage loss of activity which occurs in a sample in 2.0 hours.

[4]

- 2 (a) Explain the term

- (i) *laminar flow*,
- (ii) *incompressible flow*.

[2]

- (b) (i) Derive the equation of continuity for the flow of an ideal fluid.

- (ii) Fig. 2.1 shows oil of density 800 kg/m^3 flowing in a tube.

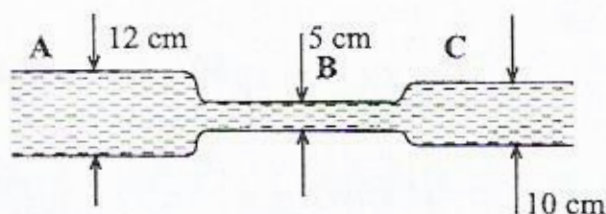


Fig. 2.1

The flow rate of oil at C is $5 \text{ cm}^3/\text{s}$. Find the velocity of the oil in section A and B.

- (iii) The pressure in Section A in Fig. 2.1 is 100 kPa.

Determine the pressure in B.

- (iv) Suggest **any** deduction that could be made from the answer in b(iii).

[10]

- 3 (a) Define *charge* and the *coulomb*. [2]
- (b) An electric kettle takes 2.5 minutes to boil 2 litres of water when it is connected to a 240 V supply.

Assume that all the electrical energy is converted into 2.4×10^5 J of heat energy.

- (i) Calculate the
1. current drawn from the supply,
 2. quantity of electric charge that flowed in the kettle for 2.5 minutes.

- (ii) The heating element in the kettle is made of nichrome wire of total length 2.4 m.

Calculate the diameter of the wire used if the resistance of nichrome is $1.1 \times 10^{-6} \Omega \text{m}$

- (iii) The diameter of the wire was found to be 0.29 mm using a micrometer screw gauge. This is different from the answer in b(ii).

Give **three** possible reasons for this observation.

[10]

- 4 (a) (i) Define a *thermometric property*.
- (ii) Give any **two** characteristics of a good thermometric property.

[3]

- (b) State properties which make a resistance thermometer

1. very accurate,
2. **not** suitable for measuring rapidly changing temperature.

[3]

- (c) A resistance thermometer has a resistance of 5.168Ω at 273.15 K and a resistance of 5.790Ω at 373.15 K.

Calculate the temperature in degrees celcius when the resistance is 4.872Ω .

[3]

- (d) Two students measured the temperature at a point, in a blast furnace, simultaneously using different thermometers. The reading on a thermocouple thermometer was 1 380 K and on a resistance thermometer was 1 390 K.

Give reasons for the different readings of the same temperature.

[3]

- 5 (a) (i) State any **three** assumptions in the derivation of the equation

$$P = \frac{1}{3} \frac{Nm \langle c^2 \rangle}{V}.$$

- (ii) Suggest the effect of the collisions of gas particles being inelastic. [6]

- (b) Fig. 5.1 shows the pressure-volume variation in an engine cylinder from intake of fuel through to release of waste gases.

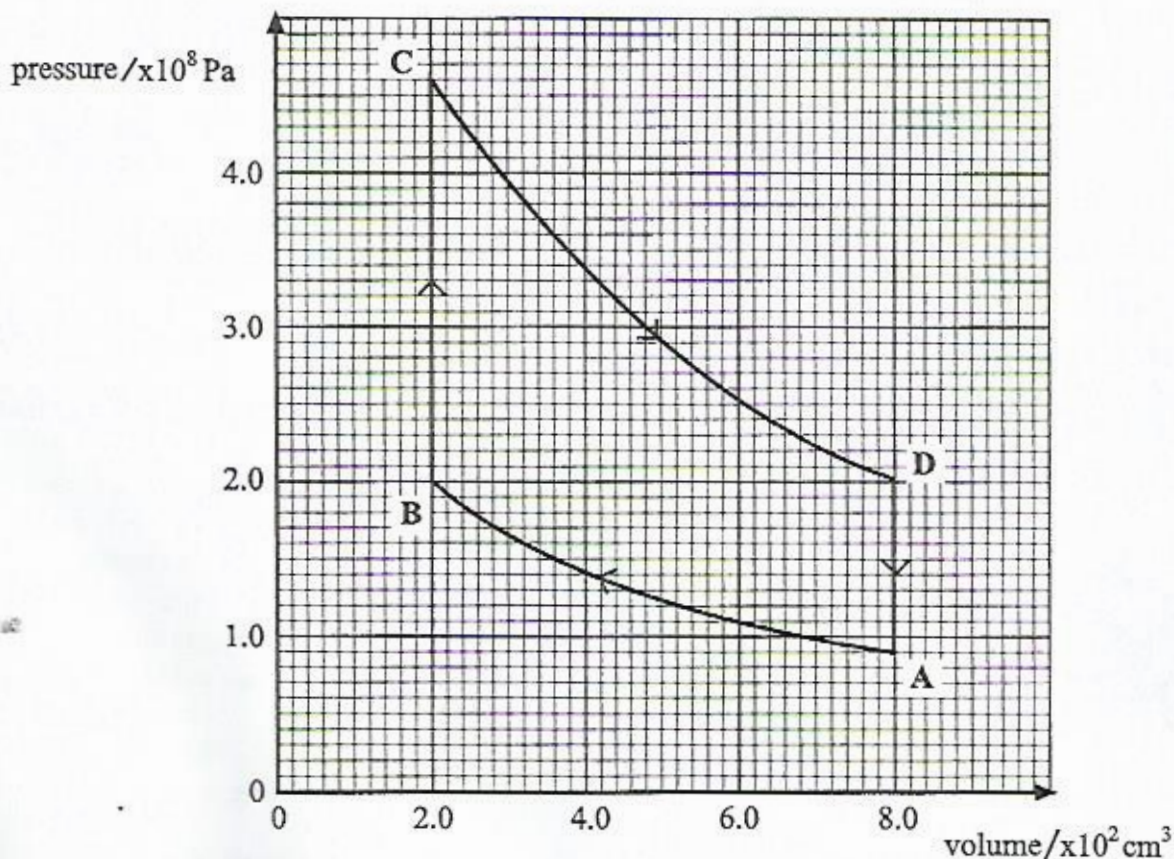


Fig. 5.1

- (i) Estimate the net work done by the cylinder piston in one cycle.
 (ii) Describe the changes occurring shown by the line BC and DA.
 (iii) Suggest how the efficiency of the system can be improved.

[6]